3.6 Reducing the miss penalty using multi-level caches

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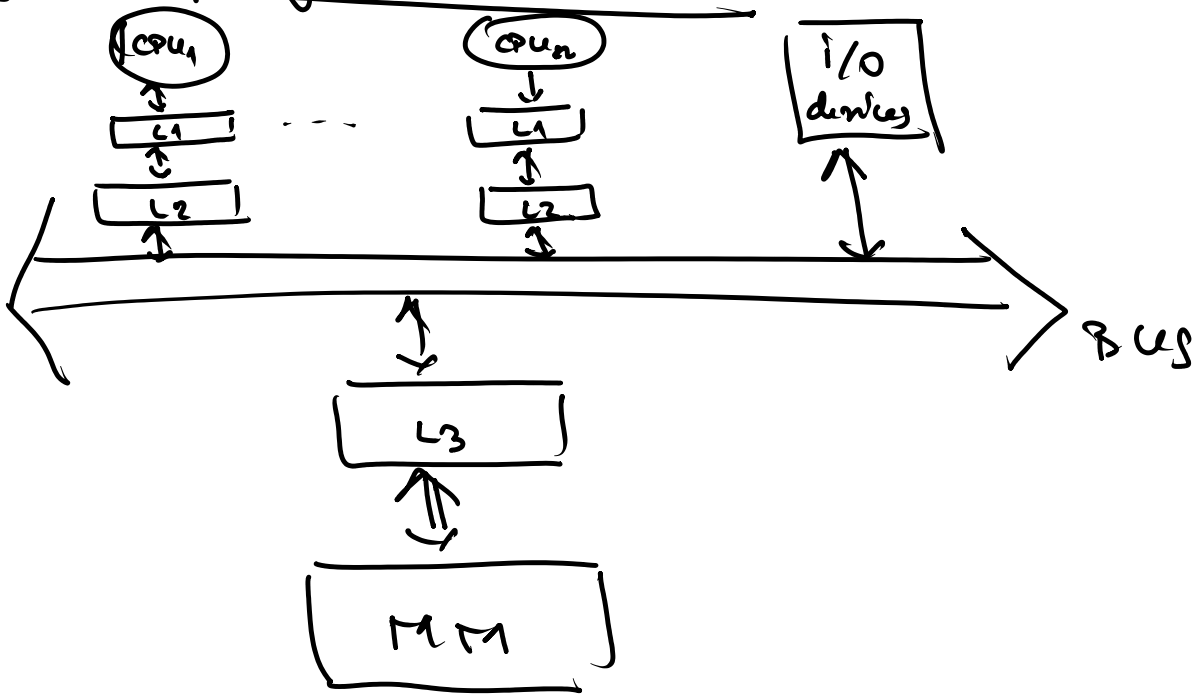

<u>Example</u>

- $CPI_{ideal} = 1.0$ (assuming all references hit in the primary cache)
- Clock rate = 4 GHz $\Rightarrow$ Clock cycle time $= \frac{1}{4 \cdot 10^9} = 0.25 \cdot 10^{-9} s = 0.25 ns$
- MM access time = 100 ns (including all miss handling)
- Miss Rate per Instruction = 2% (at the primary cache)
- How much faster would be a comp. sys with a L2 cache having 5 ns access time for either a hit or a miss

  Assume that the new comp system will have a miss rate = 0.5%

Miss Penalty to MM $= \left\lceil \frac{100 ns}{0.25 ns} \right\rceil = 400$ c.c. Miss Penalty to L2 $= \left\lceil \frac{5 ns}{0.25 ns} \right\rceil = 20$ c.c.

Total $CPI_{2\text{-level}}$ = CPI_{ideal} + $\underbrace{\text{Memory stall clock cycles per instruction}}_{\text{Misses per Instruction} \times \text{Miss Penalty}}$

$= 1 + 0.02 \times \underline{400 c.c} = 9$ c.c.

Total CPI

$$= 1 + 0.02 \times \underline{400 \text{ c.c}} = 9 \text{ c.c.}$$

$$\text{Total CPI}_{\text{2-level}} = \text{CPI ideal} + \text{Primary stalls per Instr.} + \text{Secondary stalls per Instr}$$

$$= 1 + \underbrace{0.02 \times 20 \text{ c.c}}_{0.4} + \underbrace{0.005 \times 400 \text{ c.c.}}_{2} = \boxed{3.4 \text{ c.c}}$$

$$\text{Speedup}_{\text{2-level}} = \frac{\text{CPU time}_{\text{1-level}}}{\text{CPU time}_{\text{2-level}}} = \frac{\cancel{IC} \times \text{CPI}_{\text{1-level}} \times \cancel{CCT}}{\cancel{IC} \times \text{CPI}_{\text{2-level}} \times \cancel{CCT}} = \frac{9}{3.4}$$

$$= 2.65 \checkmark$$